

EECS 245, Fall 2026

LEC 4 Simple Linear Regression

→ Read: Ch. 2

Agenda

- Modeling recipe for simple linear regression } 2.1
- Partial derivatives } 2.2
- Deriving and using optimal parameters } 2.3
- Correlation } 2.4

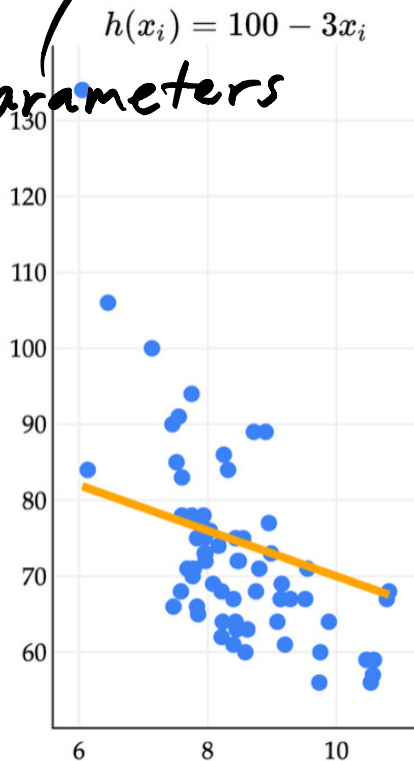
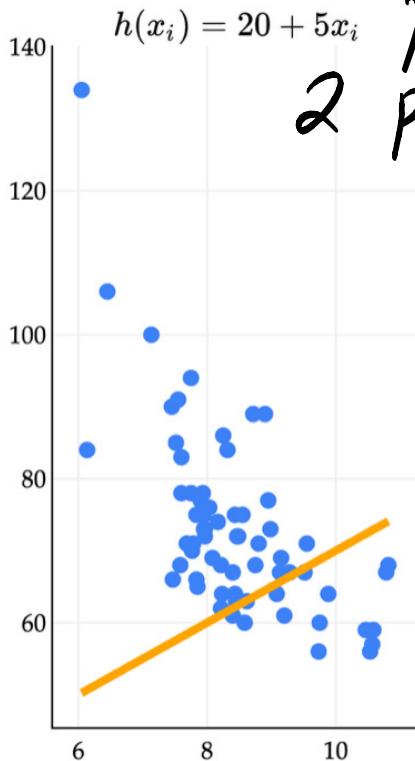
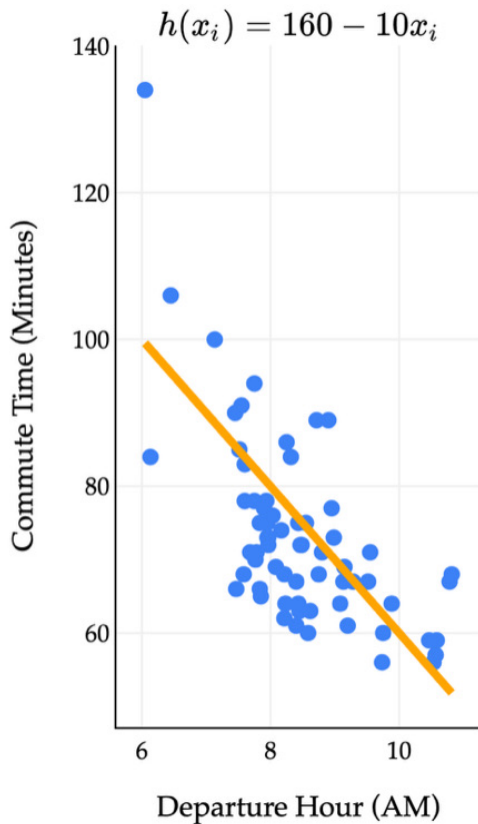
Read Ch. 2! Lots of smaller sections (not that long overall)

Announcements

- HW 1 + Welcome Survey due tomorrow
→ come to OH after lecture!
- Review Lab 2 solutions

$$h(x_i) = w_0 + w_1 x_i$$

2 parameters



Three-step modeling recipe (empirical risk minimization)

① Choose a model

$$h(x_i) = w_0 + w_1 x_i$$

simple linear regression (SLR)

② Choose a loss function

$$L_{sq}(y_i, h(x_i)) = (\overset{\text{actual}}{y_i} - \overset{\text{pred}}{h(x_i)})^2$$

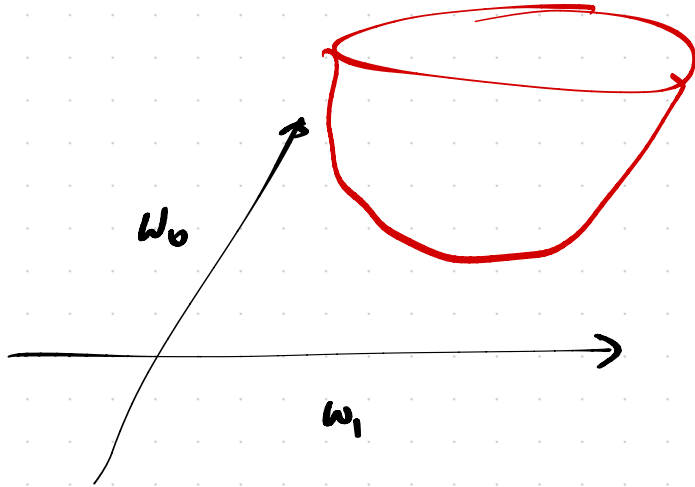
③ Minimize average loss to find optimal parameters

$$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i))^2$$

mean squared error for SLR model

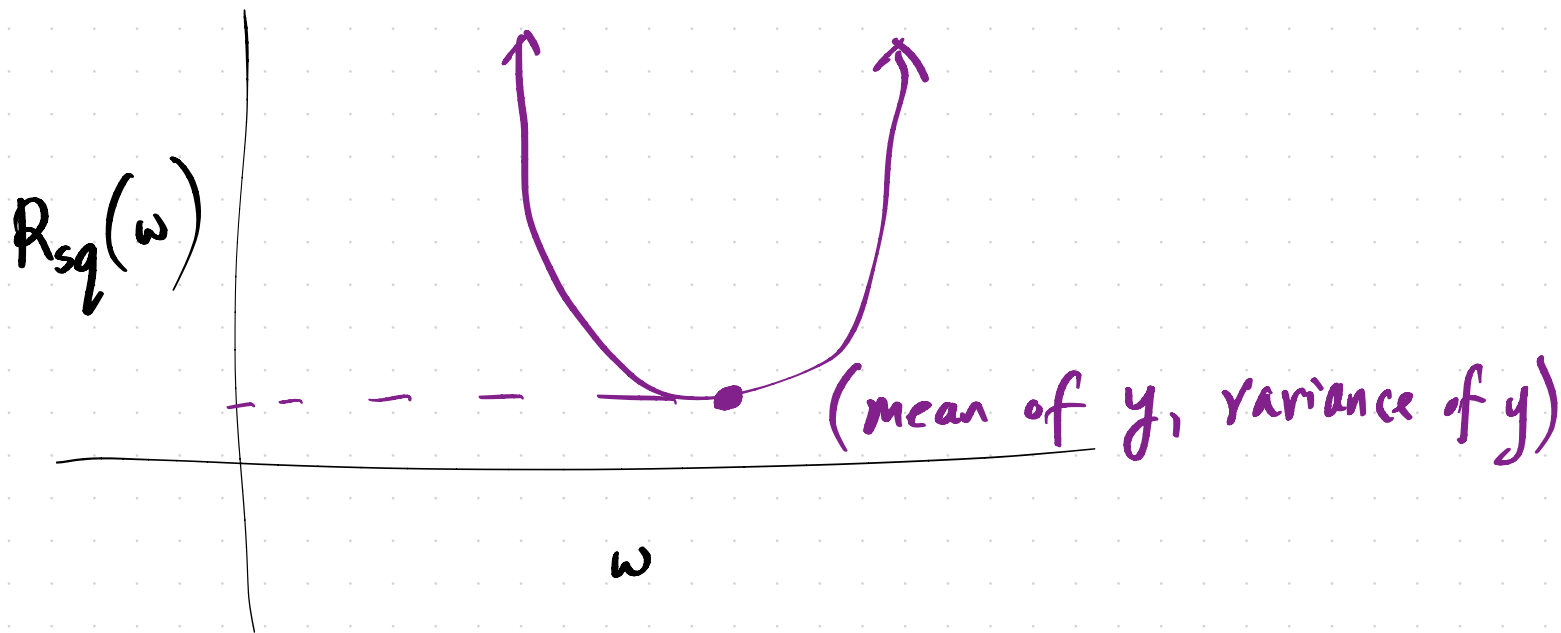
$$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n \left(y_i - \underbrace{(w_0)}_{\text{intercept}} + \underbrace{w_1 x_i}_{\text{slope}} \right)^2$$

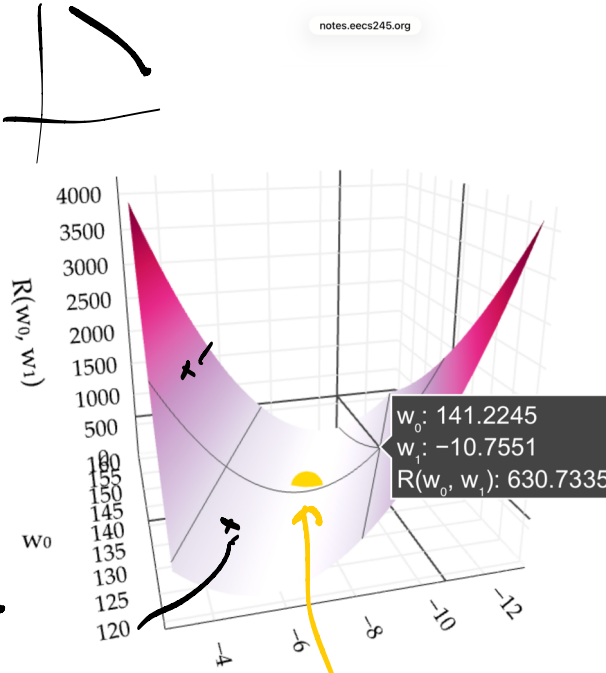
Q: what does this look like?



Recall:

$$R_{sq}(w) = \frac{1}{n} \sum_{i=1}^n (y_i - w)^2$$





go play with
loss surface
in Ch. 2.1!

(w_0^*, w_1^*)

The graph above is called a **loss surface**, even though it's a graph of empirical risk, i.e.

average loss, not the loss for a single data point. The plot is interactive, so you should drag

Detour: Partial derivative (2.2)

$$f(x, y) = \frac{x^2 + y^2}{9}$$

Suppose f is a function with multiple input variables, e.g., $f(x, y, z, \dots)$

The partial derivative of f with respect to x , $\frac{\partial f}{\partial x}$ is the derivative where all other variables are held constant.

$$f(x, y) = \frac{x^2 + y^2}{9} = \frac{x^2}{9} + \frac{y^2}{9}$$

$$\frac{\partial f}{\partial x}(x, y) = \frac{2x}{9}$$

$$\frac{\partial f}{\partial y}(x, y) = \frac{2y}{9}$$

Pick a point, e.g. $x = -3$, $y = 1/2$

$$\frac{\partial f}{\partial x}(-3, 1/2) = -\frac{6}{9} = -\frac{2}{3} \approx -0.67$$

$$\frac{\partial f}{\partial y}(-3, 1/2) = \frac{1}{9} \approx 0.11$$

$$g(x, y) = x^3 - 3xy^2 + 2 \sin(x) \cos(y)$$

$$\frac{\partial g}{\partial x}(x, y) = 3x^2 - 3y^2 + 2 \cos(x) \cos(y)$$

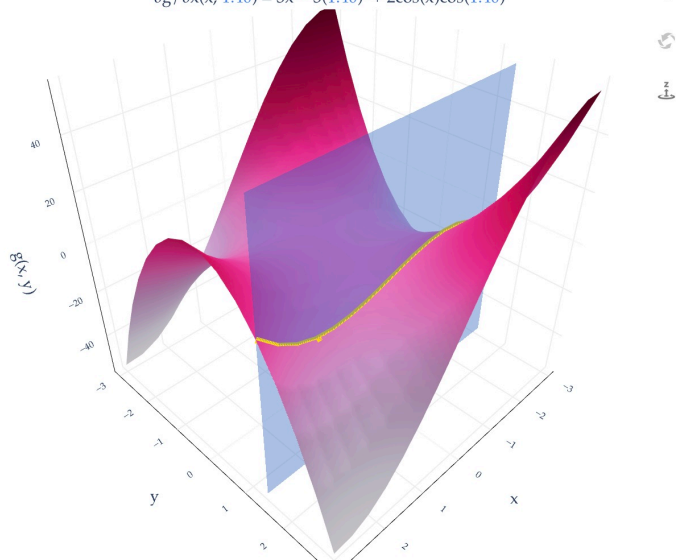
$$\frac{\partial g}{\partial y}(x, y) = -6xy - 2 \sin(x) \sin(y)$$

Important: both partial derivatives
can involve both variables!

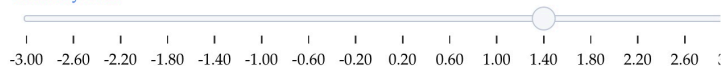
$$g(x, y) = x^3 - 3xy^2 + 2\sin(x)\cos(y)$$

$$\text{on the plane } y = 1.40, \quad g(x, 1.40) = x^3 - 3x(1.40)^2 + 2\sin(x)\cos(1.40)$$

$$\partial g / \partial x(x, 1.40) = 3x^2 - 3(1.40)^2 + 2\cos(x)\cos(1.40)$$



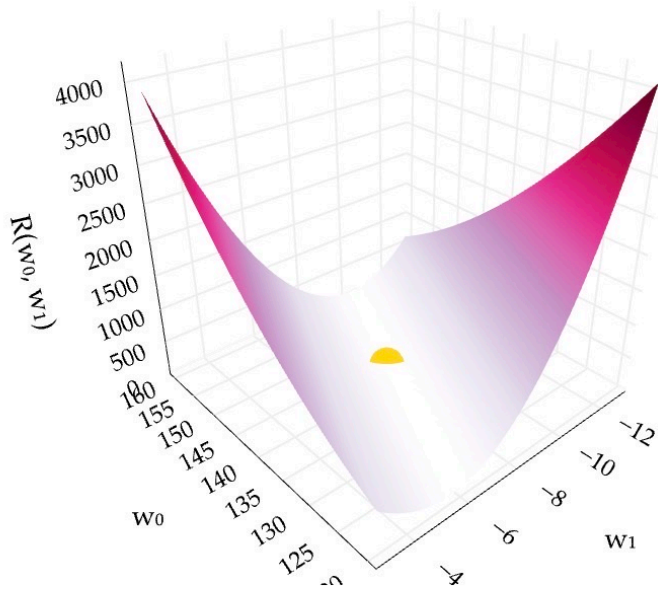
Slice at $y=1.40$



Drag the slider to $y = 1.40$ for example, and look at the **yellow curve** that results. The

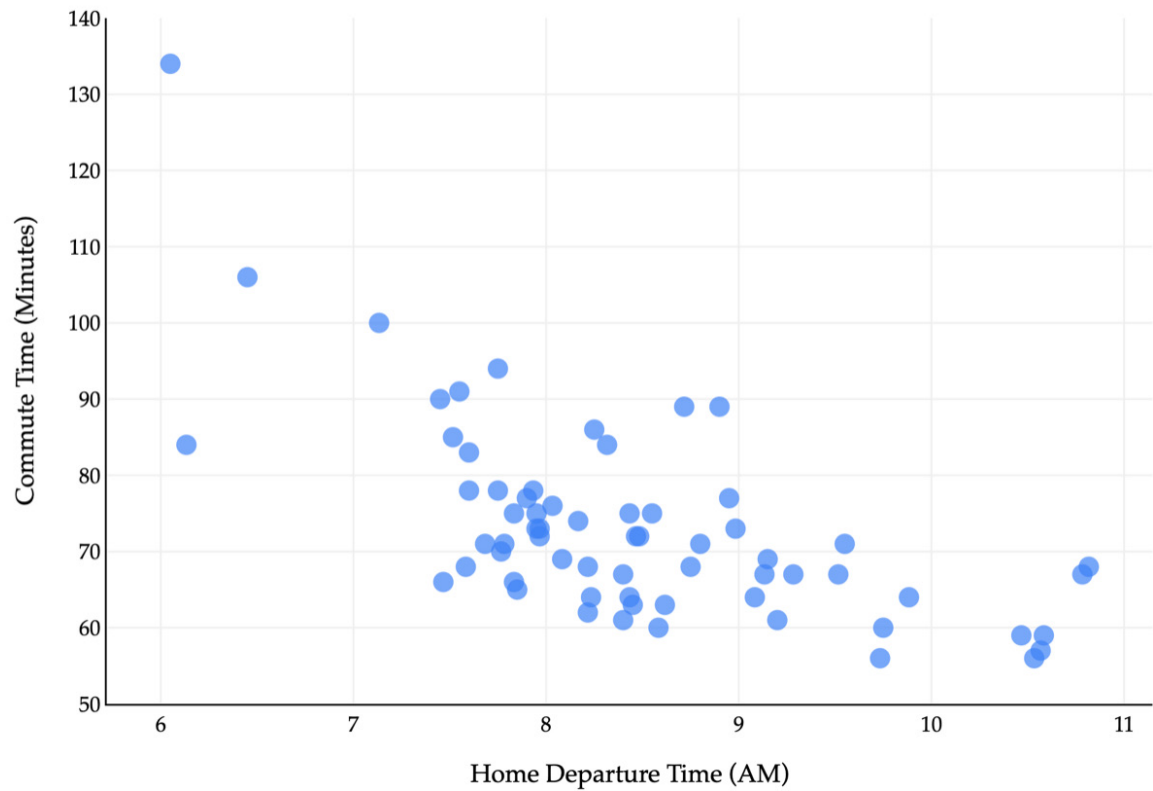
play with slider in
2.2!

$$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n \left(y_i - \underbrace{(w_0}_{\text{intercept}} + \underbrace{w_1 x_i}_{\text{slope}}) \right)^2$$



Q: How do we find w_0^* , w_1^* ?

A: Take $\frac{\partial R_{sq}}{\partial w_0}$, $\frac{\partial R_{sq}}{\partial w_1}$,
 set both = 0, and solve
 for w_0, w_1 !



$$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^n \left(y_i - \underbrace{(w_0)}_{\text{intercept}} + \underbrace{w_1 x_i}_{\text{slope}} \right)^2$$

$$\frac{\partial R(w_0, w_1)}{\partial w_0} = \frac{1}{n} \sum_{i=1}^n 2(y_i - (w_0 + w_1 x_i))(-1)$$

$$\frac{\partial R(w_0, w_1)}{\partial w_1} = \frac{1}{n} \sum_{i=1}^n 2(y_i - (w_0 + w_1 x_i))(-x_i)$$

↑ only difference!

$$\frac{\partial R}{\partial w_0} = -\frac{2}{n} \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) = 0 \quad (1)$$

$$\frac{\partial R}{\partial w_1} = -\frac{2}{n} \sum_{i=1}^n x_i (y_i - (w_0 + w_1 x_i)) = 0 \quad (2)$$

$$(1) \quad \sum_{i=1}^n \overset{\text{(actual-pred)}}{(y_i - (w_0 + w_1 x_i))} = 0$$

$$(2) \quad \sum_{i=1}^n x_i (y_i - (w_0 + w_1 x_i)) = 0$$

English interpretation:
errors must sum to 0!

$$\textcircled{1} \quad \sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) = 0$$

$$\textcircled{2} \quad \sum_{i=1}^n x_i (y_i - (w_0 + w_1 x_i)) = 0$$

$\Rightarrow$ In eq $\textcircled{1}$, solve for w_0^* , it will involve w_1

$\Rightarrow$ Then substitute into $\textcircled{2}$ to find w_1^*

$$\sum_{i=1}^n (y_i - (w_0 + w_1 x_i)) = 0$$

$$\sum_{i=1}^n y_i - \sum_{i=1}^n w_0 - \sum_{i=1}^n w_1 x_i = 0$$

$$\sum_{i=1}^n y_i - n w_0 - w_1 \sum_{i=1}^n x_i = 0$$

$$\frac{\sum_{i=1}^n y_i}{n} - w_1 \frac{\sum_{i=1}^n x_i}{n} = w_0$$

$\Rightarrow w_0^* = \bar{y} - w_1^* \bar{x}$
best intercept depends
on best slope!

Idea: substitute $w_0^* = \bar{y} - w_1^* \bar{x}$ into

$$(2) \quad \sum_{i=1}^n x_i (y_i - (w_0 + w_1 x_i)) = 0$$

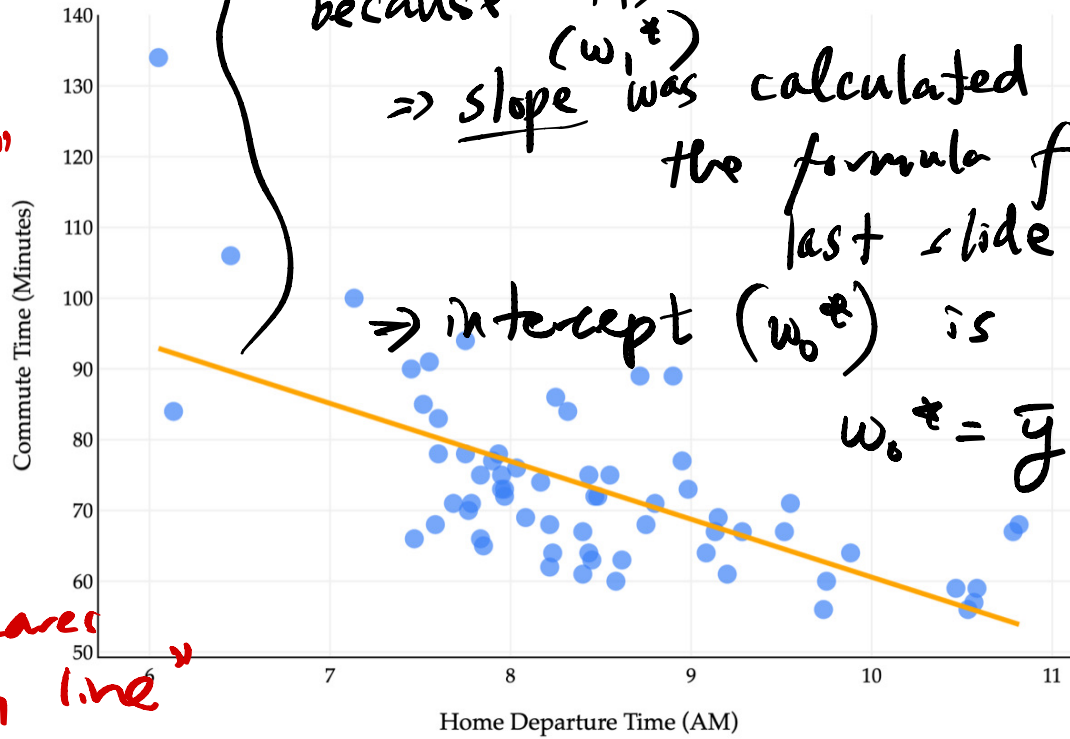
only unknown is w_1^*

$$w_1^* = \frac{\sum_{i=1}^n x_i (y_i - \bar{y})}{\sum_{i=1}^n x_i (x_i - \bar{x})}$$

"line of best fit"

"the regression line"

"least squares regression line"



This line minimizes mean squared error because its (w_1^*) slope was calculated with the formula from the last slide

$\Rightarrow$ intercept (w_0^*) is

$$w_0^* = \bar{y} - w_1^* \bar{x}$$

Several equivalent formulas for w_1^*

$$w_1^* = \frac{\sum_{i=1}^n x_i (y_i - \bar{y})}{\sum_{i=1}^n x_i (x_i - \bar{x})}$$

ALL EQUAL!

$$\begin{aligned} &= \frac{\sum_{i=1}^n x_i (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ &= \frac{\sum y_i (x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \end{aligned}$$

Key fact

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x}$$

$$= \sum_{i=1}^n x_i - n \bar{x}$$

$$= n \bar{x} - n \bar{x}$$

$$= 0$$

$$\sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= \sum x_i (y_i - \bar{y}) - \sum_{i=1}^n \bar{x} (y_i - \bar{y})$$

$$= \sum x_i (y_i - \bar{y}) - \bar{x} \sum (y_i - \bar{y})$$

$$= \sum x_i (y_i - \bar{y})$$